

Cover letter: *Mechanism-class clinical track record, not target affinity, predicts cognition-drug repurposing success*

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Submission to *Nature Communications*

Dear Editors,

I am submitting “**Mechanism-class clinical track record, not target affinity, predicts cognition-drug repurposing success**” for consideration. The contribution is a leakage-audited **negative result**: on adjudicated pivotal-trial outcomes for 31 cognition drugs, no a-priori target-centric predictor (binding affinity (MAMMAL), target genetics (Open Targets), knowledge-graph network propagation (PrimeKG), nor their ensemble) clears chance, while a drug’s **mechanism-class clinical history** separates SUCCESS from FAILURE completely. The result is not merely retrospective: **two pre-registered class predictions have already read out as predicted** (iclepertin and luvadaxistat, both NMDA-coagonist-enhancement failures flagged in advance from the bitopertin precedent) with two falsifiable predictions still pending (zatoilmilast / PDE4 and KarXT / M1-M4). This is the unusual case of a methodology paper with verified prospective evidence already on record.

Cognition-drug repurposing is class-aware, not affinity-driven — and the prior is already predicting live trials

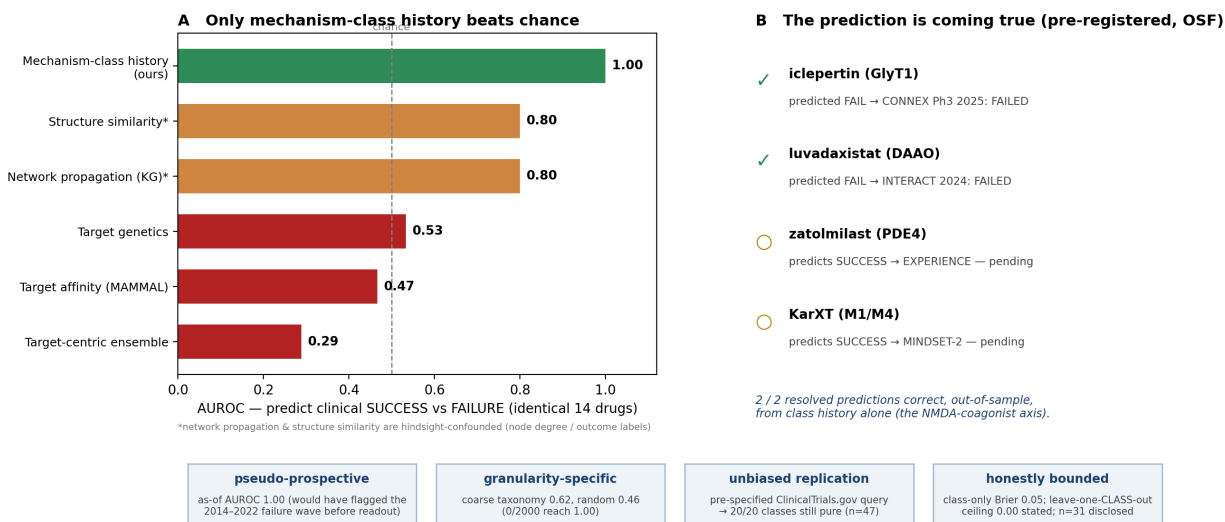


Figure 1: Graphical abstract. (A) On the identical drugs, only mechanism-class history beats chance; the two comparators that appear to (network propagation, structure similarity) are hindsight-confounded. (B) The class prior is already predicting live trials: two pre-registered predictions confirmed at readout, two pending. (C) The result is pseudo-prospective, granularity-specific, unbiased-replicated, and honestly bounded.

Why this matters. Computational repurposing for cognition is dominated by two paradigms: rank candidates by predicted target-binding affinity (increasingly with protein-ligand foundation models such as MAMMAL) and by target genetic evidence. Both encode a reasonable prior, engage a relevant target, but neither was built to answer the operative question: *will a drug of this kind actually improve cognition in patients?* Tested directly against pivotal-trial outcomes, they do not clear chance; a knowledge-graph network-propagation score and a chemical-structure score appear to, but only through hindsight a leakage audit exposes (node degree, outcome labels). This is a concrete, generalisable caution for the knowledge-graph and foundation-model repurposing literature: the kind of methodological correction that travels well beyond cognition.

The rigor behind the headline. Beyond the leakage audit and an exact label-permutation test (0/5000), the result is *pseudo-prospective*: a prequential “as-of” analysis predicting each drug from only strictly-earlier drugs reaches AUROC 1.00 and would have flagged the 2014–2022 cognition-trial graveyard before it read out; *granularity-specific* (a coarser taxonomy collapses the signal to 0.62 and random class labels to 0.46 (0/2000 reaching 1.00)); *calibrated*) a leave-one-compound-out predictor has Brier 0.05, and the target-centric features add nothing; and *robust to unbiased sourcing*: drawing trials from a pre-specified ClinicalTrials.gov query (a 294-trial denominator) and adjudicating outcome-blind leaves 20/20 mechanism classes outcome-pure.

What I am not claiming. The ledger is modest ($n = 31$; a curated expansion to 42 and an unbiased-sourced replication to 47 preserve the pattern) and outcome-pure by mechanism class, so the perfect AUROC is descriptive of class homogeneity, not a generalisable oracle, and the method cannot forecast a genuinely novel mechanism (the leave-one-class-out ceiling of 0.00 is stated beside every headline). The definitive remaining steps (a complete results-level adjudication of the full ClinicalTrials.gov denominator, an independent second ledger, and the resolution of the pending prospective predictions) are named in the paper rather than substituted for with scale not yet in hand. The work is reproducible on a single consumer GPU; all code, curated data, figures, a continuous-integration test suite, and an OSF pre-registration are public under Apache-2.0.

Fit for *Nature Communications*. The paper pairs a methodologically rigorous negative result of broad relevance to computational drug discovery (applicable to any foundation-model or knowledge-graph repurposing pipeline) with an unusually direct, already-partly-confirmed prospective commitment. That combination of methods significance and clinical-prediction interest matches the journal’s computational-biology and drug-discovery scope, and the honesty-forward framing (pre-registration, leakage audit, named limitations) fits the journal’s data- and code-availability standards.

Suggested reviewers. Marinka Zitnik (Harvard; PrimeKG and knowledge-graph repurposing (informed on the network-propagation comparator); David Ochoa (EMBL-EBI; Open Targets) the genetics paradigm evaluated); Brian M. Schilder (Imperial College London; computational repurposing methods and reproducibility); and Lon S. Schneider (USC; Alzheimer’s clinical-trial methodology, a clinician who knows the cognition-trial failure landscape and can vet the outcome ledger). **Opposed reviewers:** authors of the specific target-centric pipelines this work evaluates as underperforming (sequence-only DTI foundation-model and target-popularity / KG-degree repurposing methods) have a potential conflict and are respectfully requested to be excluded; they would review fairly, but the conflict is real.

I believe this is a useful and unusually honest contribution: a negative result that reorients how foundation models and knowledge graphs should be used in cognition repurposing, defended by a forward commitment already half-confirmed. I would be grateful to have it considered.

With thanks,

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